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Grade: Check Plus

DC27 – STATISTICS & DATA ANALYSIS

ASSIGNMENT 2

1. $\mu = \Sigma x f(x) = (20 \cdot 0.05) + (21 \cdot 0.06) + (22 \cdot 0.07) + (23 \cdot 0.08) + (24 \cdot 0.09) + (25 \cdot 0.12)$
$+ (26 \cdot 0.15) + (27 \cdot 0.13) + (28 \cdot 0.10) + (29 \cdot 0.10) + (30 \cdot 0.05) \boxed{= 25.41}$

✓ Q1

$\sigma^2 = \Sigma (x-\mu)^2 f(x) = (20-25.41)^2(0.05) + (21-25.41)^2(0.06) + \ldots + (30-25.41)^2(0.05)$
$= (29.2681 \cdot 0.05) + (19.4481 \cdot 0.06) + \ldots + (21.0681 \cdot 0.05) = 7.501892$

$\sigma = \sqrt{7.501892} \boxed{= 2.74}$

2a. $P(25.41 - 2.74 \le V \le 25.41 + 2.74) = P(22.67 \le V \le 28.15)$
$= P(23) + P(24) + P(25) + P(26) + P(27) + P(28)$
$= 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10$
$\boxed{= 0.67 \text{ or } 67\%}$

✓ Q2a

2b. $P(25.41 - 2 \cdot 2.74 \le V \le 25.41 + 2 \cdot 2.74) = P(19.93 \le V \le 30.89)$
$= P(20) + P(21) + \ldots + P(30)$
$= 0.05 + 0.06 + \ldots + 0.05$
$\boxed{= 1.0 \text{ or } 100\%}$

✓ Q2b

3. $V = \text{Price} - 25$ (with a floor of zero)

Value (v)	Probability ($g(v) = P[V=v]$)
0	0.47
1	0.15
2	0.13
3	0.10
4	0.10
5	0.05

✓ Q3

4. $P(V > 0) = P(1) + P(2) + P(3) + P(4) + P(5)$
$= 0.15 + 0.13 + 0.10 + 0.10 + 0.05$
$\boxed{= 0.53}$

✓ Q4

5. $\mu = \Sigma v g(v) = (0 \cdot 0.47) + (1 \cdot 0.15) + (2 \cdot 0.13) + (3 \cdot 0.10) + (4 \cdot 0.10) + (5 \cdot 0.05)$
$\boxed{= 1.36}$

✓ Q5

$\sigma^2 = \Sigma (v-\mu)^2 g(v) = (0-1.36)^2 \cdot 0.47 + (1-1.36)^2 \cdot 0.15 + (2-1.36)^2 + 0.13$
$+ (3-1.36)^2 \cdot 0.10 + (4-1.36)^2 \cdot 0.10 + (5-1.36)^2 \cdot 0.05 = 2.5704$

$\sigma = \sqrt{2.5704} \boxed{= 1.60}$